Spectroscopic Signature of r-process Nucleosynthesis in a Gamma-Ray Burst: Resonant Deuterium Absorption in GRB 210704A

Yuan-Chuan Zou [1,*] Xin-Ying Song [2,3,†] Qing Liu [2,3,4], Yuan-Yuan Zuo [1], Wei-Tian Deng [1,‡] and Bo-Qiang Ma [5,4§]

[1] *School of Physics, Huazhong University of Science and Technology, Luoyu Road, Wuhan, 430074, Hubei, China.*
[2] *University of Chinese Academy of Sciences, Chinese Academy of Sciences, Beijing 100049, China*
[3] *Key Laboratory of Particle Astrophysics, Institute of High Energy Physics, Chinese Academy of Sciences, Beijing 100049, China*
[4] *School of Physics, Peking University, 100871, Beijing, China and*
[5] *School of Physics, Zhengzhou University, 450001, Zhengzhou, China;*

(Dated: July 19, 2026)

Catastrophic stellar endpoint events such as binary neutron star mergers and collapsars are proposed to be sites of rapid neutron-capture process (r-process) nucleosynthesis. We report the detection of a statistically significant spectral gap between sub-GeV and GeV photons in the prompt emission of GRB 210704A. By constraining the bulk Lorentz factor, we map this gap to an energy of around $3.5 - 6.7$ MeV in the comoving frame. We rule out alternative explanations and show that this feature is naturally explained by resonant photodissociation of deuterium (^{2}H), whereas the Giant Dipole Resonance of heavier nuclei occurs at much higher energies (~ 20 MeV). Furthermore, the extremely short dynamical timescale of the prompt emission favors the survival of deuterium, as the formation of heavier r-process elements is bottlenecked by slower β-decays. This observation provides the first spectroscopic evidence of light r-process element formation within a relativistic jet, linking the immediate aftermath of the relativistic outflow generated in catastrophic stellar endpoints to nucleosynthesis.

Introduction.—Core-collapse supernovae (CC SNe) and binary neutron star (BNS) mergers both create neutron-rich environments; they are the two leading candidates for producing most of the rapid neutron-capture process (r-process) elements in the universe [e.g. 1–10]. Although the gravitational wave event GW170817 and its associated kilonova AT2017gfo confirmed that such mergers synthesize heavy elements [8, 11], observational evidence has largely been restricted to the thermal emission of radioactive decay on timescales of days. Gamma-ray bursts (GRBs) launch relativistic jets that may entrain neutron-rich material from the central engine; high-entropy winds and fireballs associated with GRBs have been proposed to be a source of light-element synthesis [12–17]. If nuclear species survive in the jet, they imprint absorption features on the gamma-ray spectrum. However, identifying these features requires precise knowledge of the bulk Lorentz factor (Γ) to translate the observed energies to the comoving frame. In this Letter, we present the discovery of a discrete spectral gap in the high-energy emission of GRB 210704A. We demonstrate that this anomaly corresponds to the photodissociation threshold of deuterium (^{2}H) in the jet's comoving frame.

Observation and Spectral Anomaly.—GRB 210704A was detected by the *Fermi* Gamma-ray Burst Monitor (GBM) and the Large Area Telescope (LAT) [18, 19]. The unusual stellar progenitor of GRB 210704A has been debated in previous work [18, 20]. The recent detailed multiwavelength analysis of GRB 210704A determines that the most likely redshift is $z = 2.34$ [21], by confirming Ly-α absorption through a line-stacking analysis of the afterglow spectrum. In the following analysis, we use $z = 2.34$ to estimate the Lorentz factor and to determine the energetics in the comoving frame.

The burst exhibits a distinct separation in its high-energy photon distribution. While the MeV emission is bright, the GeV emission ($>$ 100 MeV) arrives almost simultaneously and, crucially, displays a "zone of avoidance" in the energy spectrum.

As shown in FIG. 1, the photons are densely clustered in the sub-GeV regime (< 0.62 GeV) and the multi-GeV regime (> 1.21 GeV), leaving a statistically significant gap (denoted in gray shadow) between approximately 0.62 GeV and 1.21 GeV during the prompt phase (from 0.7 s to 2.0 s after trigger in GBM).

A full likelihood analysis is performed with GTLIKE to quantify the statistical significance of this gap. Similarly to [22], we compared a continuous power-law (PL) model with a model explicitly incorporating a spectral gap [23], and the analysis yields a preference for the gap model with $\Delta\mathrm{BIC} = -2\Delta \log(\mathrm{likelihood}) = 9.2$. This value constitutes strong statistical evidence against a simple continuous spectrum [24], further supporting the physical separation of the two emission components. [25] Furthermore, a time-resolved analysis reveals that the upper energy envelope of the sub-GeV photons and the lower energy envelope of GeV photons evolve synchronously, maintaining the gap structure. This strong correlation ($\rho > 0.7$, Spearman rank; see the details in Appendix A) suggests that the two components are physically linked but separated by an absorption mechanism.

Comoving Frame Energetics.—To interpret the physical origin of the gap, we must determine the Lorentz factor Γ. A combined model of a multicolor blackbody [mBB, e.g. see 26–28] plus a PL function for the broad energy band is used to describe the spectrum of the prompt emission from the keV to GeV energy band; with the lowest BIC and a difference of $\Delta\mathrm{BIC} = 8.1$, it is statistically preferred over the other models (see details in Figure 3 in Appendix B).

With $z = 2.34$, the Lorentz factor is estimated to be $\gtrsim 500$ around the peak flux from the photon with the highest energy

* zouyc@hust.edu.cn
† songxy@ihep.ac.cn
‡ dengwt@hust.edu.cn
§ mabq@pku.edu.cn

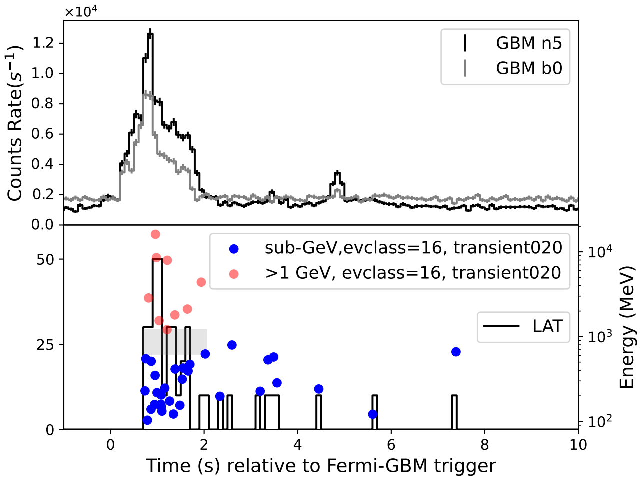

FIG. 1. (Color online) Left Y-axis with the unit of counts rate (s^{-1}): the histograms are the light curves of GBM detectors (NaI 5 and BGO 0, upper plot) and LAT (bottom plot); the high-energy photons are from the LAT data of TRANSIENT020 class (or evclass=16), and used in the analysis to maximize the statistics for spectral analysis of GRB prompt emission. Right Y-axis with the unit of MeV in bottom plot: energies of high-energy photons (denoted by circles). The gray shadow in the first 2 s denotes intersection of all gaps between sub-GeV (blue circles) and GeV (red circles) photons.

(16.09 GeV); the model-dependent method using the quasi-thermal component in the prompt phase proposes a range of 500 to 900 (see the details in Appendix C).

Applying the Doppler correction $E' \approx E_{\rm obs}(1+z)/2\Gamma$ with $\Gamma = 300$, the observed gap ([0.62, 1.21] GeV) translates to a comoving energy range of:

$$E'_{\rm gap} \approx 3.5 - 6.7\ \rm MeV. \tag{1}$$

Even with $\Gamma = 900$, the range is $1.2 - 2.2$ MeV. This energy range is critical.

Before attributing the gap to absorption by matter, we examine whether it could arise from intrinsic spectral evolution or from the superposition of different emission components, e.g. a cutoff of the prompt keV–GeV component at sub-GeV energies due to internal photon-photon absorption, with an early-afterglow component emerging at higher energies. First, the optical depth for pair production between a high-energy photon and the target MeV photons (see Appendix C) reaches $\tau_{\gamma\gamma} \gtrsim 1$ for a ~ 1 GeV photon emitted at a radius of $\sim 10^{11}$ cm only if $\Gamma \lesssim 170$ (a larger emission radius requires an even smaller upper limit of Γ), far below the estimated range of 500–900; thus the deficit of photons above 0.62 GeV cannot be caused by pair production. Similarly, for pair production to generate an absorption feature centered at ~ 1 GeV, a much larger Lorentz factor (e.g., $\Gamma \approx 3000$) than the estimate would be required. Second, the GeV photons are unlikely to originate from external shocks: in the external forward shock model, it takes a few seconds for the accelerated electrons to produce 100 MeV photons and a somewhat longer time to produce GeV photons [29], so that the $\sim$GeV emission should be delayed with respect to the 100 MeV emission by about half of the deceleration time; however, the observed 100 MeV and GeV emissions arrive almost simultaneously. Case studies of GRB 090902B and GRB 090510 also suggest that the external shock model cannot interpret the prompt GeV data [30, 31], and the modeling of [32] indicates that the external shock model may interpret the GeV emission after, but not during, the prompt phase. For GRB 210704A, the high-energy prompt emission traces the MeV emission well (FIG. 1), implying an internal origin; moreover, the correlation between the GeV photons and the upper energy envelope of the sub-GeV photons would be difficult to explain if the GeV photons were from external shocks. In addition, a two-component jet model (a fast spine and a slow sheath; see, e.g., An 33) fails to explain the synchronized temporal evolution of the sub-GeV and GeV components, and Klein-Nishina suppression in inverse Compton scattering would produce a high-energy cutoff rather than a discrete gap followed by a recovery in flux at > 1.21 GeV. Therefore, we turn to photon-matter interactions.

Resonant Nuclide Absorption.—We propose that the gap arises from the resonant absorption of photons by nuclei entrained in the jet. The comoving energy of ~ 4 MeV aligns remarkably well with the photodissociation threshold of deuterium:

$$\gamma + {}^2{\rm H} \rightarrow p + n. \tag{2}$$

The cross-section for photodissociation peaks typically around $4 - 5$ MeV [34], which perfectly matches our derived comoving energy $E'_{\rm gap}$.

Crucially, this energy scale distinguishes deuterium from all other potential nuclear absorbers. Heavier nuclides, particularly the Fe-group elements often invoked in supernova models, interact with photons primarily through the Giant Dipole Resonance (GDR). The GDR peak energy for nuclei with mass number A is approximately $E_{\rm GDR} \approx 31.2A^{-1/3} + 20.6A^{-1/6}$ MeV [35], which places the resonance at $15 - 25$ MeV for heavy elements (e.g., ^{56}Fe) [36]. Such absorption would manifest at observed energies > 3 GeV, far above the gap detected in GRB 210704A. The unique low-energy threshold of deuterium makes it the only viable candidate for an absorption feature at ~ 4 MeV.

If the gap indeed arises from resonant deuterium absorption, the resonance energy in turn almost fixes the Lorentz factor to be around 300; we therefore choose $\Gamma = 10^{2.5}$ as a typical value in the following.

Timescales and Nucleosynthesis. — With the standard internal shock model for the prompt emission, the emitting radius is determined to be [37]

$$R \simeq 2\Gamma^2 c\delta t/(1+z) \simeq 6 \times 10^{12}\Gamma_{2.5}^2\delta t_{-3}/(1+z)\ {\rm cm}, \tag{3}$$

where δt is the timescale of the pulse duration; hereafter, the convention $Q_s = Q/10^s$ is adopted in cgs units throughout the text. To extract δt, a wavelet decomposition is performed with the data of the internal shock emission (photons of energy $<$ 20 keV are required, and the contribution from the quasi-thermal emission is 7%; the light curve has a sampling rate of 4000 Hz; see the details of the method in MacLachlan *et al.* 38). δt is determined to be about 1 ms with an uncertainty of a few ms. Considering the redshift ($z = 2.34$), the emitting radius is $R \sim 2 \times 10^{12}$ cm.

The identification of deuterium is further supported by the dynamical timescales of the GRB jet. The dynamical time in the comoving frame is extremely short, only a fraction of a second:

$$t'_{\rm dyn} \approx \frac{R}{\Gamma c} \sim 0.1 R_{12} \Gamma_{2.5}^{-1}\ {\rm s}. \tag{4}$$

In a neutron-rich outflow ejected from the central engine, nucleosynthesis begins with the rapid capture of neutrons by protons ($p + n \to {}^2$H). To form heavier r-process elements (up to lanthanides), the chain must proceed through the creation of seed nuclei. However, the formation of seeds heavier than deuterium is rate-limited by weak interactions (β-decays), which typically require timescales of seconds to minutes to proceed significantly; a timescale of ~ 0.1 s is far too short for heavier elements to form.

Given the relativistic expansion, the jet material is "frozen out" before significant quantities of heavy elements can form via β-decay. Consequently, the jet composition at the photospheric radius is expected to be dominated by nucleons and light isotopes, specifically deuterium, rather than heavy metals. This kinetic constraint robustly reinforces the deuterium interpretation.

Optical depths for photon-electron and photon-deuteron interactions in relativistic internal shocks.— The comoving density of the jet, mainly contributed by protons (with a rest mass of m_p), is expressed in terms of the isotropic matter luminosity ($L_{\rm K} = E_{\rm K}/T_{\rm dur,lab}$; $E_{\rm K}$ is the kinetic energy and $T_{\rm dur,lab}$ is the duration time in the lab frame) of the jet at the non-thermal emission radius R as

$$n'_p = \frac{L_{\rm K}}{4\pi R^2 c \Gamma^2 m_p c^2} = 1.8 \times 10^{17} L_{\rm K,55} R_{12}^{-2} \Gamma_{2.5}^{-2}\ {\rm cm}^{-3}, \tag{5}$$

where $L_{\rm K,55} = L_{\rm K}/10^{55}\,{\rm erg\,s}^{-1}$ and the unit of R is cm. With the comoving density of deuterium n'_D and the number fraction of deuterium (Y_D), the optical depth for the photodissociation of deuterium is

$$\tau_{\gamma D} \sim \frac{\sigma_{\gamma D} n'_D R}{\Gamma} = 1.4 Y_D L_{\rm K,55} R_{12}^{-1} \Gamma_{2.5}^{-3}, \tag{6}$$

where $\sigma_{\gamma D} \simeq 2.5 \times 10^{-27}$ cm^2 at a photon energy $\varepsilon'_\gamma \simeq 4.48$ MeV in the deuteron rest frame [34].

On the other hand, the GeV photons should escape freely from the scattering off electrons. Internal shocks are natural sites for the dissipation of the kinetic energy of a baryonic fireball [39–42]. In the internal shock model of gamma-ray bursts, electrons accelerated at relativistic shocks are commonly assumed to follow a non-thermal distribution [43],

$$N(\gamma_e) \propto \gamma_e^{-p}, \qquad \gamma_e \geq \gamma_m, \tag{7}$$

where γ_m is the minimal Lorentz factor of the electrons accelerated in the internal shocks. For typical internal shock parameters adopted in the literature, $\gamma_m \sim 10^{3.5}$–10^6 [44, 45]. The scattering regime between photons and relativistic electrons is determined by

$$x \equiv \frac{\gamma_e \varepsilon'_\gamma}{m_e c^2}, \tag{8}$$

where ε'_γ is the photon energy in the comoving frame. One has $x \gtrsim 8 \times 10^5$ for $\varepsilon'_\gamma \sim 4$ MeV and $\gamma_m \simeq 10^5$ as typical values. In this Klein-Nishina limit, the total scattering cross section is suppressed to [46]

$$\sigma_{\rm KN} \simeq \frac{3}{8} \sigma_T \frac{\ln(2x) + 1/2}{x} \lesssim 4.6 \times 10^{-30}\ {\rm cm}^2, \tag{9}$$

thus, the optical depth for Compton scattering in the Klein-Nishina regime is

$$\tau_{\rm KN} \sim \frac{\sigma_{\rm KN} n'_e R}{\Gamma} \lesssim 2.6 \times 10^{-3} L_{\rm K,55} R_{12}^{-1} \Gamma_{2.5}^{-3} \ll 1, \tag{10}$$

assuming local electric neutrality in the jet, $n'_e \simeq n'_p$.

The estimates above show that the photons falling in the resonant band should be absorbed by the deuterium, while the other photons are able to escape from the scattering off the electrons.

Discussion.—Let us demonstrate the plausibility of the observation of photodissociation of deuterium in the jet of GRB 210704A as follows: the first key point is whether the baryon loading in a relativistic outflow could produce enough deuterium; second, GeV photons are emitted while those in the resonant region are absorbed by deuterium, which requires that $\tau_{\gamma D}$ and $\tau_{\rm KN}$ (for GeV photons) satisfy

$$\tau_{\rm KN} < 1 \lesssim \tau_{\gamma D}. \tag{11}$$

The total isotropic-equivalent luminosity of the outflow could be estimated from the radiated luminosity $L_{\gamma,\rm iso}$ and the radiative efficiency (ϵ_γ) as

$$L_{\rm tot,iso} = L_{\gamma,\rm iso}/\epsilon_\gamma; \tag{12}$$

for a baryonic fireball, $L_{\rm K} \lesssim (1 - \epsilon_\gamma) L_{\rm tot,iso}$. It is reasonable that $\epsilon_\gamma \lesssim 10\%$, because the radiation efficiency of internal shocks is typically low [e.g., 37, 47]. $L_{\gamma,\rm iso} = 1.0 \times 10^{54}$ erg s^{-1} for the component from the keV to GeV energy band (about 2.6 times that of the peaking component in the MeV emission); therefore, we assume that $L_{\rm K}$ is comparable to 10^{55} erg s^{-1}. In fact, $L_{\gamma,\rm iso}$ obtained from the modeling is time-averaged; the density of baryons is not uniform, and $L_{\rm K}$ for each pulse (or shell) is even larger.

Y_D can be as large as $\simeq 10\%$ under the right conditions in neutron-ion decoupling [see the simulation in 16]. A large production of deuterium could also proceed through the spallation of α-particles [17], which is caused by neutron-ion decoupling, as well as by internal shocks. Although nucleosynthesis is sensitive to the unknown details of the intense internal shocks occurring in the jet of GRB 210704A, we expect that Y_D might be even greater due to the spallation of α-particles. Considering that the fraction of α-particles could be comparable with that of nucleons for the short dynamical time [16, 17],

$Y_{\rm D} \simeq 0.3$ is not impossible and Eq. (11) is satisfied in this case. For the Lorentz factor, $\Gamma \simeq 300$ is a plausible value that could lead to $\tau_{\gamma D} \gtrsim 1$. Given $\gamma_m \simeq 5\times10^6$ and $p = 2$ for the power-law distribution of γ_e, even photons of much smaller energy, e.g. $E_{\rm obs} \simeq 10$ keV, satisfy $\tau_{\rm KN} < 1$. This is well consistent with the observed prompt spectrum.

For the high-energy photons, because of the very small optical depth for Compton scattering, the photons of the PL component escape from the outflow once produced. The interactions between the high-energy photons and their target photons with lower energies are in the relativistic regime; thus the Lorentz factor of the outgoing pairs is $\gamma_\pm \gg 1$ (where $\gamma_\pm \equiv \frac{E_0}{m_e c^2}$ and E_0 is the incoming photon energy in the center-of-momentum frame), and the cross section for pair production is suppressed to [48, 49]

$$\sigma_{\gamma\gamma} = \frac{3}{8}\sigma_{\rm T}\gamma_\pm^{-2}(\ln 2\gamma_\pm^2 - 1) \simeq 10^{-31}\ {\rm cm}^2 \ll \sigma_{\rm T}. \quad (13)$$

$\tau_{\gamma\gamma} \lesssim 1$ for the high-energy photons of 1.21 GeV−4.39 GeV, which constitute the upper boundary of the gap, requires $\Gamma \gtrsim 300$ at $R \sim 10^{12}$ cm, which is well consistent with our hypothesis.

Conclusion.—We identified a statistically significant spectral gap in the prompt emission of GRB 210704A. Interpreting this for the first time as resonant absorption by deuterium nuclei in a jet with $\Gamma \simeq 300$, we find a consistent physical picture that links the burst to a neutron-rich progenitor. This represents the first direct spectroscopic detection of the ingredients of r-process nucleosynthesis in the prompt gamma-ray phase, opening a new window for studying nuclear astrophysics in extreme environments.

Previous studies of GRB 210704A noted its peculiar properties, despite its duration lying in the intermediate range [18, 20, 21]. Unlike kilonovae, which probe the sub-relativistic ejecta days after the event, this spectral feature probes the relativistic jet itself seconds after launch, revealing the initial chemical composition of the r-process flow. It should be noted that such observations are expected to be rare: GRB 210704A is a 'short' long burst that favors the abundance of deuterium, and the high baryon loading of the outflow together with the rapid variability of the central engine creates the conditions for the occurrence of intense internal shocks.

ACKNOWLEDGEMENTS

The authors are very grateful for the public GRB data from Fermi/GBM and Fermi/LAT. Dr. X.Y. Song thanks the support of the National Natural Science Foundation of China (grant No.12303052). Q. Liu and Dr. X.Y. Song are very grateful for the support from China's Space Origins Exploration Program. Prof. B.Q. Ma and Q. Liu appreciate the support of the National Natural Science Foundation of China (grant No.12335006). The authors thank anonymous reviewers for their useful comments. The authors thank Prof. Shuang-Nan Zhang, Prof. Ming-Yu Ge, and Prof. Francesco Longo for useful comments and discussions on the data analysis.

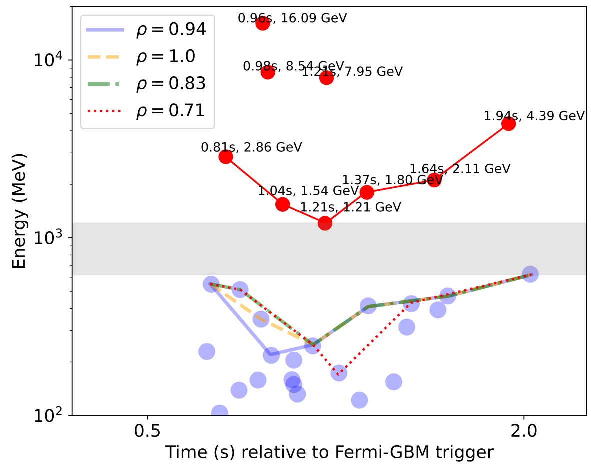

FIG. 2. The Spearman rank correlation coefficients of temporal evolution between $\gamma_{\rm GeV}^{\rm envelope}$ and $\gamma_{\rm sub-GeV}^{\rm envelope}$ (denoted by different line styles and colors) extracted with different bin-edge division ratios. The gray shadow between 0.62 and 1.21 GeV shows the intersection of all gaps between sub-GeV and GeV photons.

Appendix A: The possible correlation between $\gamma_{\rm sub-GeV}^{\rm envelope}$ and $\gamma_{\rm GeV}^{\rm envelope}$

The Spearman rank correlation coefficient (ρ) [50][51] is used to quantify the monotonic relationship of the temporal evolution between the energy of $\gamma_{\rm GeV}^{\rm envelope}$ (denotes the lower envelope of GeV photons) and that of $\gamma_{\rm sub-GeV}^{\rm envelope}$ (denotes the upper energy envelope of sub-GeV photons). $\gamma_{\rm GeV}^{\rm envelope}$ is naturally formed by the GeV photons as shown in FIG. 2 in red circles connected by a red line. In order to ensure a one-to-one correspondence between $\gamma_{\rm sub-GeV}^{\rm envelope}$ and $\gamma_{\rm GeV}^{\rm envelope}$, the maximum sub-GeV energy value in each time bin was extracted as $\gamma_{\rm sub-GeV}^{\rm envelope}$; the data binning in the range of $T_0 + [0.70, 2.05]$ s is performed as follows: 1) the edges of the bin are determined to be the n-equal division points in time of each adjacent GeV point in $\gamma_{\rm GeV}^{\rm envelope}$; 2) to reduce the impact of artificial selection, the bin-edge division ratio (rate) is tested from 0.01 to 0.99 with an increment of 0.01 (for example, a rate equal to 0.2 means left five-division point).

From all binning results with different bin-edge division ratios, four different energy envelopes of sub-GeV photons were extracted, as shown in FIG. 2. ρ is determined to be > 0.7 conservatively from the results of all sets of bin edges, which implies a strong correlation between $\gamma_{\rm sub-GeV}^{\rm envelope}$ and $\gamma_{\rm GeV}^{\rm envelope}$[52].

Appendix B: Spectral modelings with different models and the analysis of the prompt emission

FIG. 3 shows the time-integrated spectral modeling results with mBB + PL, BAND + PL and mBB + BAND. The parameters of the three best models are listed below (BIC and χ^2 over the degrees of freedom are shown in the figures):

- BAND+PL: for the BAND component, $\alpha = -0.06^{+0.14}_{-0.11}$, $\beta = -2.80^{+0.26}_{-0.35}$, $E_{\rm p} = 256.1^{+13.4}_{-16.1}$ keV, $F_{\rm BAND} = 10.7^{+0.7}_{-0.7} \times 10^{-6}$ erg cm^{-2} s^{-1}; for the PL component: index $p = -1.80^{+0.18}_{-0.09}$, $F_{\rm PL} = 20.2^{+3.7}_{-4.2} \times 10^{-6}$ erg cm^{-2} s^{-1};
- mBB+PL: for the mBB component, $m = -0.29^{+0.37}_{-0.37}$, $kT_{\rm min} = 19.1^{+3.0}_{-3.5}$ keV, $kT_{\rm max} = 146.0^{+25.2}_{-17.3}$ keV, $F_{\rm mBB} = 8.5^{+0.4}_{-0.3} \times 10^{-6}$ erg cm^{-2} s^{-1}; for the PL component: $p = -1.81^{+0.06}_{-0.03}$, $F_{\rm PL} = 22.4^{+5.6}_{-5.6} \times 10^{-6}$ erg cm^{-2} s^{-1};
- mBB+BAND: for the mBB component, $m = -0.37^{+0.55}_{-0.60}$, $kT_{\rm min} = 18.8^{+3.7}_{-5.3}$ keV, $kT_{\rm max} = 145.1^{+20.0}_{-14.1}$ keV, $F_{\rm mBB} = 7.7^{+0.4}_{-0.4} \times 10^{-6}$ erg cm^{-2} s^{-1}; for the BAND component: $\alpha = -1.68^{+0.04}_{-0.02}$, $\beta = -1.90^{+0.16}_{-0.07}$, $\log_{10}(E_{\rm p}/{\rm keV}) = 3.89^{+0.17}_{-0.14}$, $F_{\rm BAND} = 23.3^{+1.7}_{-1.8} \times 10^{-6}$ erg cm^{-2} s^{-1}.

From the residual distribution from 40 keV to 300 keV, there exists a systematic bias in the modeling with the BAND + PL functions. The change in BIC is ΔBIC = 8.1, which shows a strong preference for mBB + PL. Therefore, the mBB function is better than the BAND function for describing the MeV peak component from the modeling results. In addition, the low-energy photon index for the peaking component is about -0.1, although it cannot be decisive evidence for the quasi-thermal emission. The ΔBIC of mBB+BAND relative to mBB+PL is 1.8; although BAND is not an unreasonable model for the component of the broad energy band, the mBB + BAND model is not better than mBB + PL.

It is evident that there exist two different components in the prompt emission. There are two candidate processes to interpret the observation:

1) There exist seed photons in the outflow, and some of them become the component of a broad energy band from keV to GeV via inverse Compton (IC) scattering by relativistic electrons. However, not all of them are emitted, especially those with lower energies, which correspond to a larger scattering cross section with electrons; moreover, as the electrons cool, the cross section of electron-photon scattering becomes larger and even increases to the Thomson cross section ($\sigma_{\rm T} > \sigma_{\rm KN}$). They will be thermalized below the photosphere and emitted when the radius reaches $R_{\rm ph}$; $R_{\rm ph}$ satisfies (r_0 is the initial radius for jet launching, and ranges from 10^7 cm to 10^9 cm) [49]

$$R_{\rm ph} \simeq (\frac{L_{\rm K}\sigma_{\rm e\gamma}}{8\pi m_p c^3 \Gamma})^{1/3} r_0^{2/3} = 4.9 \times 10^{11} L_{\rm K,55}^{1/3} (\frac{\sigma_{\rm e\gamma}}{\sigma_{\rm T}})^{1/3} \Gamma_{2.7}^{-1/3} r_{0,8}^{2/3} \text{ cm} \quad \text{(B1)}$$

for the unsaturated acceleration regime; for the saturated acceleration regime,

$$R_{\rm ph} \simeq \frac{L_{\rm K}\sigma_{\rm e\gamma}}{8\pi m_p c^3 \Gamma^3} = 4.7 \times 10^{13} L_{\rm K,55} (\frac{\sigma_{\rm e\gamma}}{\sigma_{\rm T}}) \Gamma_{2.7}^{-3} \text{ cm}. \quad \text{(B2)}$$

The observed time delay of this component relative to the keV–GeV component, $\simeq (1+z)R_{\rm ph}/(2c\Gamma^2)$, is less than a few milliseconds for $R_{\rm ph} \lesssim 10^{11} - 10^{13}$ cm. This means that the two components arrive almost simultaneously, which is consistent with the observation. (We have also used the cross-correlation function to obtain the time lag between the emission below 20 keV, where the contamination from the quasi-thermal component is less than 7%, and that in 50–100 keV; the lag is determined to be around 0. Although the prompt emission is a superposition of the two components, this implies that there is no very large time delay between them.)

2) The synchrotron self-Compton (SSC) process in the internal shocks could also account for the two components in the prompt phase. The peaking MeV component comes from synchrotron emission of the accelerated electrons, while some of them behave as seed photons for the IC scattering. However, the peaking MeV emission should not be quasi-thermal since $\tau_{\rm KN} < 1$, which is not consistent with the modeling results that the mBB function is better than the BAND function in describing the peak MeV emission. Secondly, given $L_{\gamma,\rm peak} \simeq 4 \times 10^{53}$ erg s^{-1} in $T_0 + [0.7, 2.05]$ s, similar fractions of the electron and magnetic-field energies, and $E_{\rm p} \sim 250$ keV, the typical Lorentz factor of the emitting electrons is estimated following [37]; $\gamma_{e,\rm p} \gtrsim 10^{3.5}$ requires the peaking component to be emitted at a large radius ($R \sim 10^{14}$ cm) so that $\tau_{\rm KN}(E_{\rm obs} \sim 250\,{\rm keV}) < 1$ is satisfied, whereas at such a large radius the optical depth for the photodissociation of the deuteron would be much less than 1.

Thus, we think that the first explanation is more consistent with the observations. In addition, because most of the energy in the prompt phase is contributed by the keV–GeV component ($L_{\gamma,\rm keV-GeV} = 1.0 \times 10^{54}$ erg s^{-1}, about 2.6 times that of the peaking component, $L_{\gamma,\rm peak} \simeq 3.7 \times 10^{53}$ erg s^{-1}), and $L_{\rm K}$ is estimated from this component, the emission mechanism of the MeV peaking component does not affect the calculation and conclusion much.

Appendix C: The estimates of Γ

1. Estimate of the Range of Γ with quasi-thermal emission

The approach for the estimate of Γ is sketched in broad terms in the typical fireball model of GRBs [e.g. 53, 54] as follows: the thermal component originates from the photosphere of an expanding plasma jet; the observed thermal flux (integrated over all frequencies) is given by integrating the intensity over the emitting surface [55],

$$F_{\rm BB}^{\rm obs} = \frac{2\pi}{d_L^2} \int_{\mu_{\rm min}}^{\mu_{\rm max}} d\mu \mu r_{ph}^2 \mathcal{D}^4 (\sigma_{\rm SB} T_{\rm ph}'^4/\pi). \quad \text{(C1)}$$

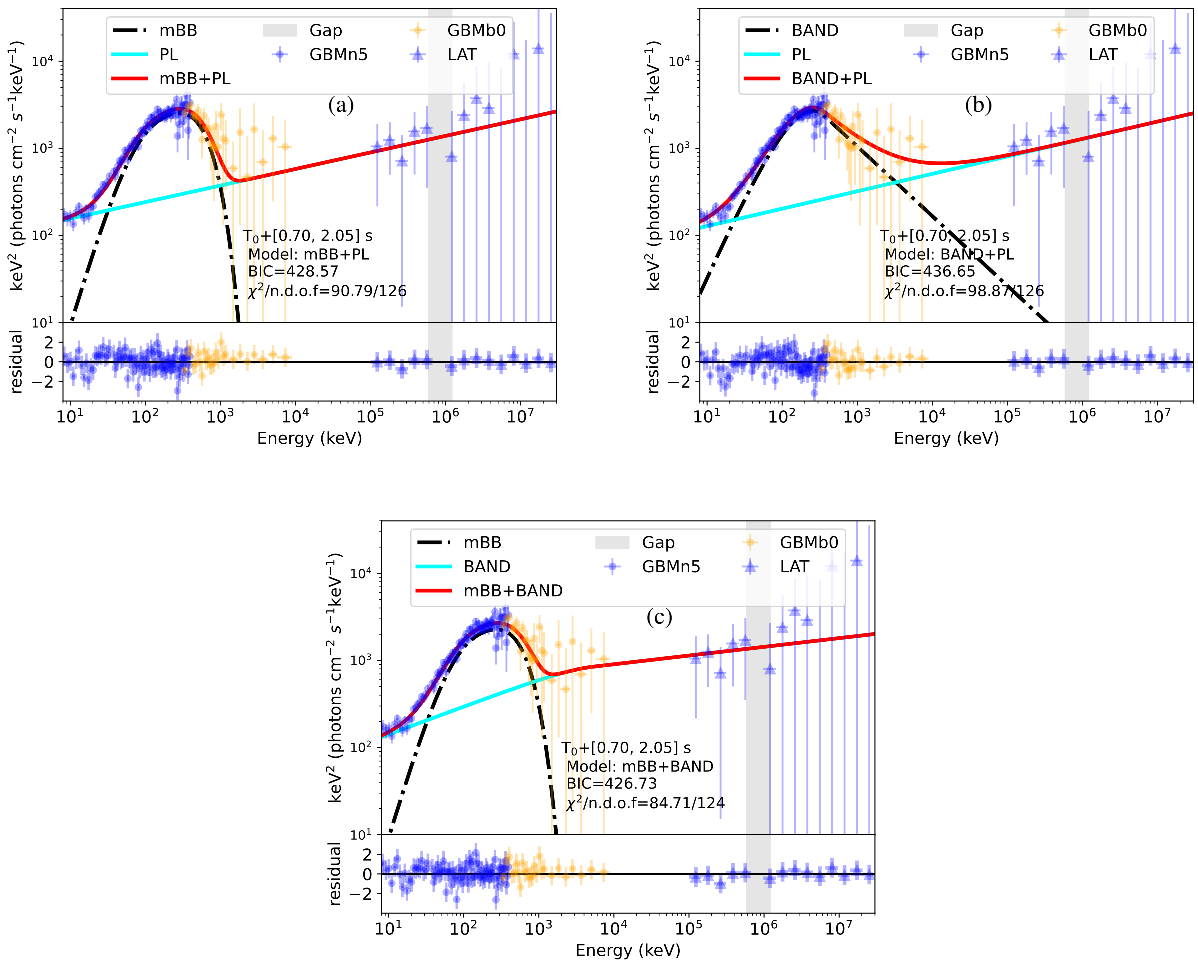

FIG. 3. (Color online) Modeling results with (a) mBB+PL, (b) BAND+PL and (c) mBB+BAND; the bins inside the gap (gray) are excluded from the fits.

Here, $T'_{\rm ph}$ is the comoving temperature at the photospheric radius, $\sigma_{\rm SB}$ is Stefan's constant and $\mathcal{D} = \mathcal{D}(\theta)$ is the Doppler factor, $\mathcal{D} = [\Gamma(1-\beta\mu)]^{-1}$. The angle θ is the angle between the direction of the outflow velocity vector (β) and the line of sight, $\mu \equiv \cos(\theta)$, and $\Gamma = (1-\beta^2)^{-1/2}$ is the outflow Lorentz factor. The observed radiation is dominated by photons emitted close to the line of sight ($\theta_{\rm min} = 0$), the upper integration boundary in the equation for the thermal flux is $\mu_{\rm max} = \cos\theta_{\rm min} = 1$, thus one has [55, 56]:

$$T_{\rm obs} = C_1 \Gamma_{\rm ph} T'_{\rm ph}/(1+z), \tag{C2}$$

$$F^{\rm obs}_{\rm BB} = \mathcal{R}^2 \sigma_{\rm SB} T^4_{\rm obs}, \tag{C3}$$

where

$$\mathcal{R} = C_2 \, \frac{(1+z)^2}{d_L} \frac{r_{\rm ph}}{\Gamma_{\rm ph}}; \tag{C4}$$

$C_1 \simeq 1.48$ and $C_2 \simeq 1.06$ are factors derived from the detailed numerical integration of the angle- and distance-dependent photosphere emission.

Taking into account different regimes of acceleration of the outflow, for example, in the case of saturation regime (i.e. $r_{\rm ph}$ is greater than $r_{\rm s}$, and r_s is the saturation radius), $r_{\rm ph} = L\sigma_{\rm T}/8\pi\Gamma^3 m_p c^3$ (here $L = 4\pi d_{\rm L}^2 Y F^{\rm obs}$, $Y \geq 1$ is the ratio between the total energy and the energy emitted in γ-rays, $F^{\rm obs}$ is the sum of thermal and non-thermal component; $\Gamma = \eta$, where η is the dimensionless entropy), one has

$$\Gamma = \left[(1.06)(1+z)^2 d_L \frac{Y F^{\rm obs} \sigma_T}{2 m_p c^3 \mathcal{R}}\right]^{1/4}. \tag{C5}$$

More realistically, the observed spectra are broader than the Planck spectrum, because of geometrical broadening [e.g. 26]. This means that the observed spectrum is a superposition of a series of blackbodies of different temperatures arising from different angles to the line of sight; therefore, a multi-blackbody (mBB) function could be used to describe the quasi-thermal spectrum, and the temperatures of the individual Planck function are related by a power law. In addition, the other component of the outflow (i.e., Poynting flux) also affects the estimate of Γ. Therefore, this approach is extended based on basic physical models (the so-called 'top-down' approach; see the details in Gao and Zhang 56 and Song *et al.* 57). Note that in the unsaturated regime, Γ and $r_{\rm ph}$ are related to the initial radius r_0; physically $r_0 = 10^7 - 10^9$ cm, from which the range of the Γ value is estimated.

2. Estimate of the lower limit of Γ with the photon with the highest energy

The optical depth for pair production of a high-energy photon (with observed energy of $E_{\rm GeV,obs}$) in the jet with the Lorentz factor Γ at a radius $R_{\gamma\gamma}$ is defined as

$$\tau_{\gamma\gamma} \simeq \sigma_{\gamma\gamma} n^{'}_{\gamma,>E_{\pm,\rm obs}} R_{\gamma\gamma}/\Gamma, \tag{C6}$$

where $\sigma_{\gamma\gamma}$ ($\sim 10^{-25}$ cm^2) is the cross section of $\gamma\gamma \to e^+ + e^-$, and $n^{'}_{\gamma,>E_{\pm,\rm obs}}$ is the number density of the target photons in the comoving frame; the observed energy $E_{\pm,\rm obs}$ of these target photons should satisfy the following criterion (note that an on-axis observation is assumed),

$$E_{\pm,\rm obs} > \left(\frac{2\Gamma m_e c^2}{1+z}\right)^2 \frac{1}{E_{\rm GeV,obs}}. \tag{C7}$$

The distribution of photons in the MeV-energy band could be described by $f(E_{\rm obs})$ (with a normalization factor K and a unit of keV^{-1} cm^{-2} s^{-1}). We have

$$4\pi d_{\rm L}^2 \int_{E_{\pm,\rm obs}}^{\infty} K f(E) dE = 4\pi R_{\gamma\gamma}^2 c n_{\gamma,>E_{\pm,\rm obs}} (1+z), \tag{C8}$$

where in the frame of the observer $n_{\gamma,>E_{\pm,\rm obs}} = 2\Gamma n^{'}_{\gamma,>E_{\pm,\rm obs}}$, and $R_{\gamma\gamma}$ could be given by

$$R_{\gamma\gamma} \simeq R_{\rm ph} + 2c\Gamma^2 \Delta t/(1+z), \tag{C9}$$

where Δt is the arrival time delay of high-energy photons relative to the MeV emission. With Eqs. (C6)-(C9), the requirement $\tau_{\gamma\gamma} < 1$ for the emission of the photon with the highest energy (16.09 GeV) around the peak flux provides a lower limit of $\Gamma \gtrsim 500$ at $R_{\gamma\gamma} \sim 10^{11}$ cm. Note that in this method the estimated Γ is approximately proportional to $(1+z)$, so the comoving energies are almost unaffected by the choice of the redshift.

3. The estimate of Γ in GRB 210704A

The range of Γ is estimated in different regimes using the 'top-down' approach. Regime II ($r_{\rm ra} < r_{\rm ph} < r_{\rm c}$, where $r_{\rm ra}$ is the rapid acceleration radius, and $r_{\rm c}$ is the coasting radius) is allowed. For $T_0 + [0.70, 2.05]$ s and assuming $Y = 10$, Γ ranges from 500 to 900. Note that from Eq. (C5), $\Gamma \propto Y^{1/4}$; the value of Y does not affect the estimate of Γ much, e.g., the factor of $Y^{1/4}$ for $Y = 10$ is 1.5 times that for $Y = 2$. In fact, Γ determined from the modeling with a specific model (e.g., a top-hat afterglow model; see the details in Pieterse *et al.* 21) is about 1000 ± 600, which means that the result of this approach is consistent with the model-dependent estimation from the afterglow.

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